

Task-03 Introduction and Networking basics for Cybersecurity

1.1 Introduction to Networking for Cybersecurity

Computer networking is the process of connecting devices so they can communicate and exchange data with each other over a network. Every time we browse a website, send a message, or access an online service, network communication happens in the background.

From a cybersecurity point of view, **networking is where most security risks begin**. Whenever data travels between systems, it can be intercepted, altered, or misused if not properly protected. That is why understanding how network communication works is important as it helps us see **how attacks happen, how data leaks occur, and where security controls should be placed**.

1.2 Why Networking Knowledge Is Important in Cybersecurity

Networking knowledge is important in cybersecurity because **almost every cyber-attack uses the network as its medium**. Attackers exploit weaknesses in network communication to gain access, steal data, or disrupt services.

Understanding networking helps security professionals:

- Detect suspicious or malicious traffic
- Identify unauthorized access attempts
- Understand how attacks like sniffing, spoofing, and MITM work
- Secure communication using encryption and secure protocols
- Configure firewalls and monitoring systems effectively

In simple words:

Cybersecurity protects systems, but networking shows how attackers reach those systems.

1.3 Basic Networking Concepts

1. IP Address (Internet Protocol Address)

An IP address is like a digital identity for a device on a network. It tells the network *where data should be delivered* and *where it came from*.

For example, in this task, 10.0.2.15 represents the Kali Linux machine inside the VirtualBox network, while Google's servers have public IP addresses. From a cybersecurity point of view, IP addresses help identify **attack sources, target systems, and suspicious network behavior**.

How to read an IP address:

- Example: 10.0.2.15
- It is made up of four numbers separated by dots

- Each number ranges from 0 to 255
- Private IPs (like 10.x.x.x, 192.168.x.x) are used inside local networks
- Public IPs are used on the internet

Why it matters in cybersecurity:

- Attackers scan IP addresses to find targets
- Firewalls and security rules are often based on IP addresses
- Knowing source and destination IPs helps detect suspicious traffic

Understanding IP Address Types

- **Private IP Addresses**
 - Used inside local networks
 - Example from this task: 10.0.2.15
 - Not routable on the internet
- **Public IP Addresses**
 - Used for communication over the internet
 - Example observed: Google server IP
 - Routable and reachable globally

The following table shows commonly used private IP address ranges observed in local and organizational networks.

IP Class Range	Usage
Class A 10.0.0.0 – 10.255.255.255	Private networks
Class B 172.16.0.0 – 172.31.255.255	Private networks
Class C 192.168.0.0 – 192.168.255.255	Home/office networks

2. MAC Address (Media Access Control Address)

A MAC address is a **unique hardware identity** assigned to a network interface by the manufacturer. Unlike IP addresses, MAC addresses usually **do not change**.

In networking and cybersecurity, MAC addresses are important because they help identify **actual physical devices** on a local network, especially during **device tracking** or **network access control**.

How to read a MAC address:

- Example: 52:55:0a:00:02:02
- Written in **hexadecimal format**
- Used mainly within **local networks (LANs)**

Why it matters in cybersecurity:

- Used for device identification in local networks
- Can be spoofed by attackers
- Important in attacks like **ARP spoofing**

3. Port

Ports are like **entry points** on a device that allow different services to communicate. Each service uses a specific port number, such as:

- Port 80 for HTTP
- Port 443 for HTTPS
- Port 53 for DNS

In cybersecurity, monitoring ports helps detect **unauthorized services, malware communication, and suspicious open ports.**

How ports work:

- Example: IP: Port → 10.0.2.15:443
- Common ports:
 - 80 → HTTP
 - 443 → HTTPS
 - 53 → DNS

Why it matters in cybersecurity:

- Open ports increase attack surface
- Port scanning is a common reconnaissance technique
- Misconfigured ports can expose services

4. Protocols

Protocols are **rules** that define how data is sent and received over a network. They ensure that communication between devices happens in a **standard and understandable way.**

Common protocols like TCP, UDP, DNS, HTTP, and HTTPS were observed in this task. Understanding protocols is important in cybersecurity because attackers often **abuse protocol behavior** to steal data or gain access.

Simple way to understand it:

- Protocols are like **rules of conversation**
- Both sender and receiver must follow the same rules

Why it matters in cybersecurity:

- Some protocols send data in plain text
- Others encrypt data for security
- Choosing secure protocols is critical to protect data

5. Packet

Packets are **small pieces of data** into which information is broken before being sent over a network. Each packet carries source and destination information along with the actual data.

In cybersecurity, analyzing packets helps detect **data leaks**, **malicious traffic**, and **unusual communication patterns**, which is why packet analysis tools like Wireshark are widely used.

How packets work:

- Each packet contains:
 - Source IP
 - Destination IP
 - Protocol information
 - Actual data

Why it matters in cybersecurity:

- Packets can be captured and analysed
- Attackers may inspect or manipulate packets
- Packet analysis helps detect malicious behaviour

1.4 Key Network Protocols Observed in This Task

a) ICMP (Internet Control Message Protocol)

ICMP is used mainly for **network testing and diagnostics**, such as checking whether a system is reachable.

In this task, ICMP traffic was **directly observed in Wireshark** while using the ping google.com command, showing echo request and echo reply packets.

Example in this task:

- ping google.com
- ICMP Echo Request and Echo Reply

Security relevance:

- Helps test if a system is reachable
- Can be abused for reconnaissance or flooding attacks

b) DNS (Domain Name System)

DNS translates **domain names into IP addresses**, allowing users to access websites using readable names instead of numbers.

DNS traffic was **directly observed in Wireshark** while resolving google.com, where DNS queries and responses showed the mapping between the domain name and Google's IP address.

Example:

- google.com → 142.250.183.238

Security relevance:

- DNS traffic reveals browsing activity
- Can be targeted by spoofing or poisoning attacks
- DNS monitoring helps detect suspicious domains

c) TCP (Transmission Control Protocol)

TCP is a **reliable, connection-oriented protocol** that ensures data is delivered correctly and in order.

The TCP three-way handshake was **observed in Wireshark** while loading websites, showing SYN, SYN-ACK, and ACK packets during connection establishment.

Key feature:

- Uses a **three-way handshake** (SYN → SYN-ACK → ACK)

Security relevance:

- Many attacks exploit TCP connections
- Understanding TCP helps identify scans and connection abuse

d) HTTP (Hypertext Transfer Protocol)

HTTP is used for **unencrypted web communication**, meaning the data can be read in plain text.

HTTP traffic was **directly observed in Wireshark** while visiting the neverssl.com website, where request and response data was visible in readable form.

Security relevance:

- Data can be intercepted and read
- Credentials and sensitive information are exposed

e) HTTPS (HTTP Secure)

HTTPS is the secure version of HTTP and uses **encryption** to protect data during transmission.

HTTPS traffic was **directly observed in Wireshark** while accessing google.com, where the data appeared as encrypted TLS packets instead of readable content.

What changes in Wireshark:

- Content becomes unreadable
- Only encrypted application data is visible

Security relevance:

- Protects confidentiality and integrity of data
- Essential for secure web communication

2.1 Environment Setup

The practical analysis for this task was performed using the following setup:

- **Operating System:** Kali Linux (Virtual Machine)
- **Tool Used:** Wireshark
- **Network Interface:** eth0

Wireshark was used to capture and analyze live network traffic generated during normal activities such as pinging a server, resolving domain names, and accessing websites.

2.2 ICMP Traffic Analysis (Ping)

ICMP traffic was analyzed to understand basic network connectivity and how packets travel between a client and a remote server.

Action Performed

- The command `ping google.com` was executed in the terminal.
- Wireshark was used to capture ICMP packets generated during this process.

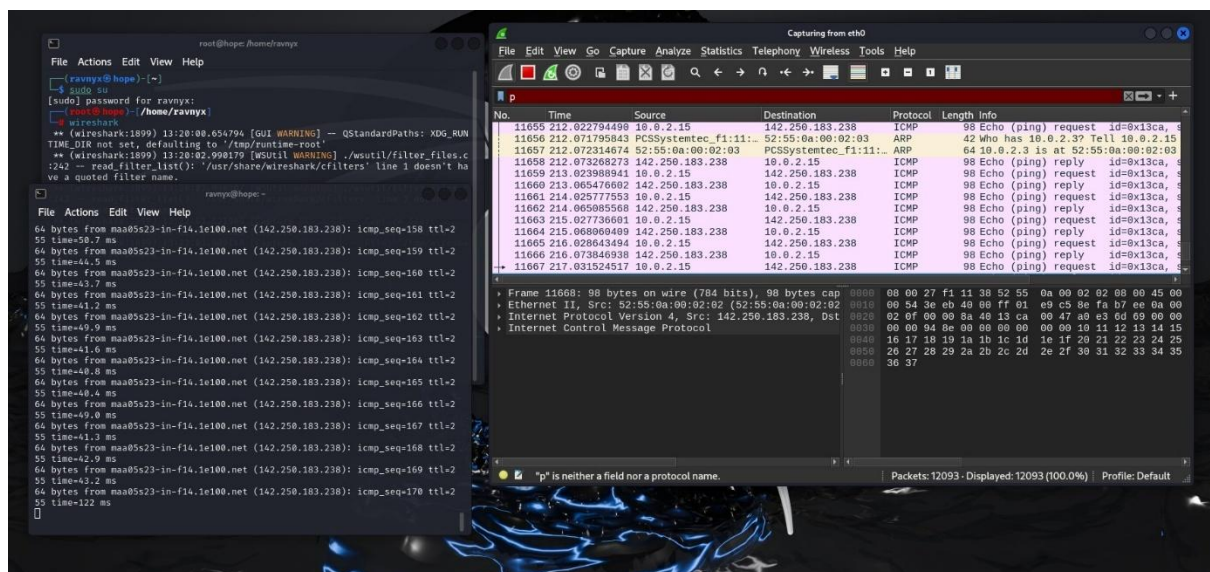


Figure 1: ICMP Echo Request and Echo Reply Packets Captured in Wireshark

Observation

- ICMP Echo Request packets were sent from the local system to Google's server.
- ICMP Echo Reply packets were received in response, confirming successful network connectivity.
- The repeating request–reply pattern shows how ICMP is used for reachability testing.

- This demonstrates how basic connectivity can be verified at the network layer.

2.3 DNS Traffic Analysis

DNS traffic was analyzed to observe how domain names are translated into IP addresses before communication begins.

Action Performed

- The command `nslookup google.com` was executed in the terminal.
- The Wireshark display filter `dns` was applied to capture DNS packets.

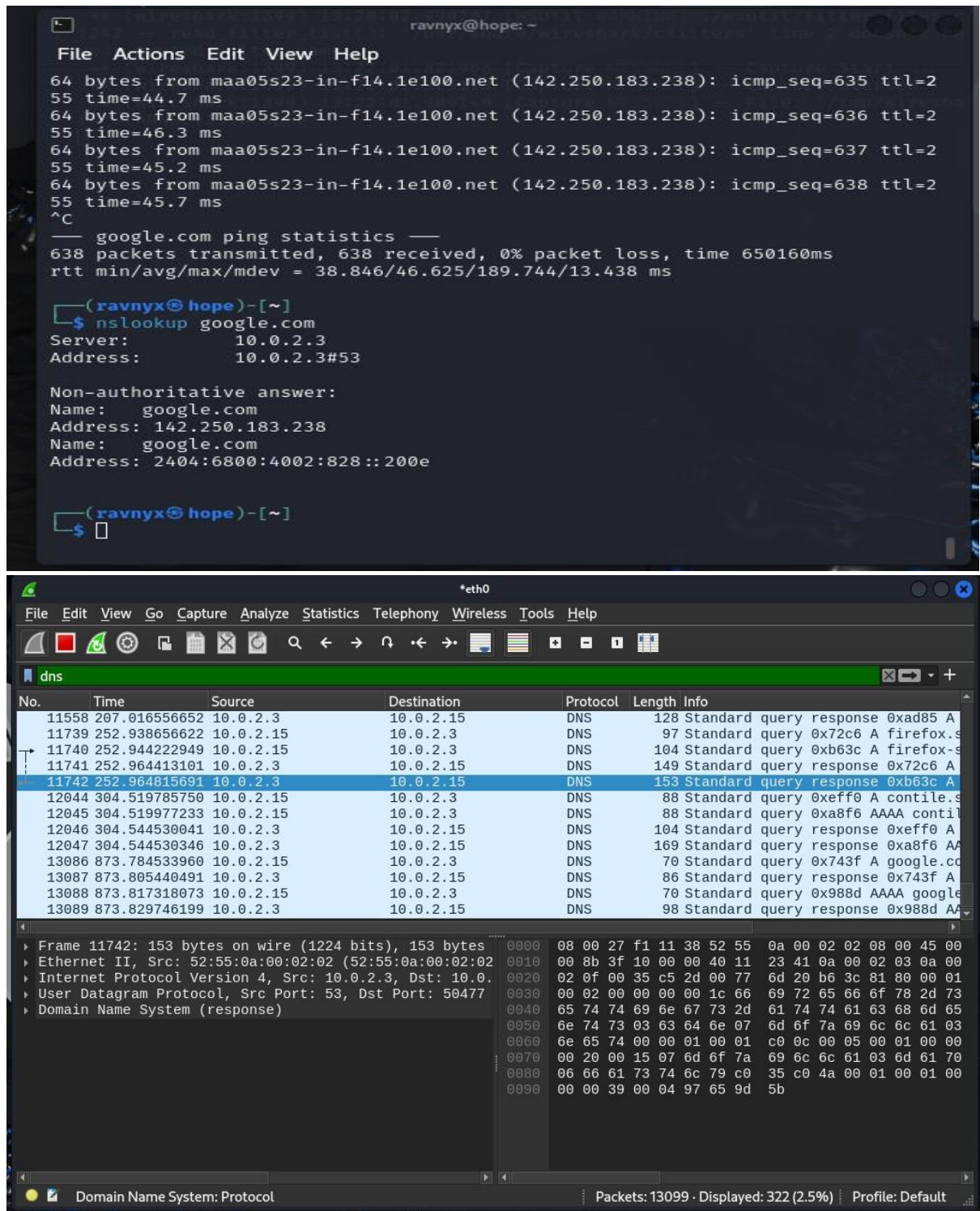


Figure 2: DNS Query and Response Captured Using Wireshark

Observation

- A DNS query was sent requesting the IP address of google.com.
- The DNS response returned both IPv4 and IPv6 addresses.

- DNS traffic was observed over UDP port 53.
- This shows that DNS resolution is the first step before any actual data exchange occurs.

2.4 HTTP Traffic Analysis (Unencrypted Communication)

HTTP traffic was analyzed to understand how data is transmitted without encryption.

Action Performed

- The website <http://neverssl.com> was accessed using the browser.
- The Wireshark display filter `http` was applied.

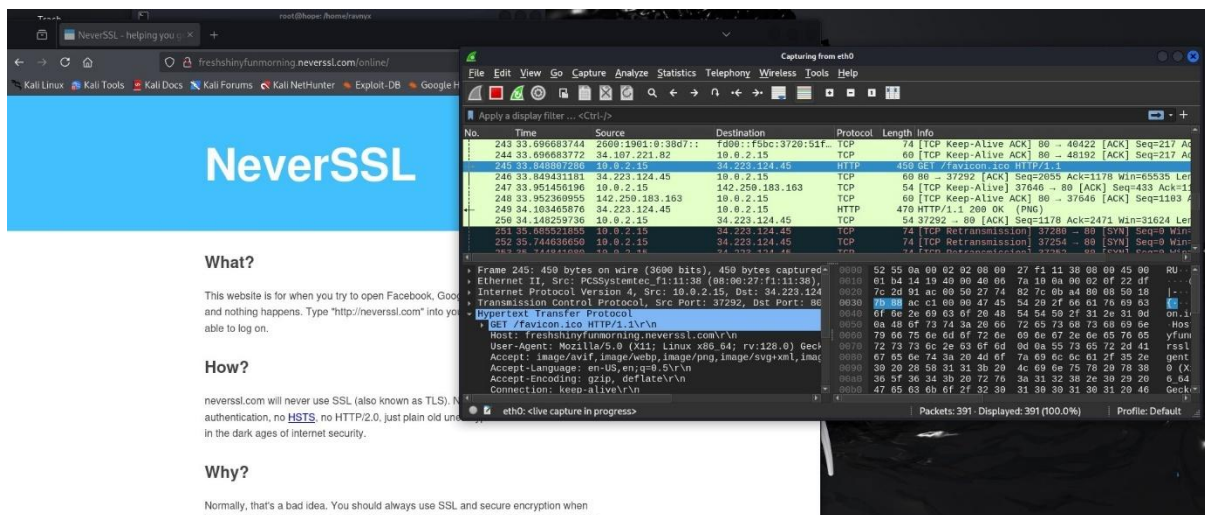


Figure 3: HTTP Traffic Showing Plaintext Request Headers

Observation

- HTTP request headers such as GET, Host, and User-Agent were visible in plaintext.
- The transmitted data could be read directly in Wireshark.
- This demonstrates that HTTP does not provide confidentiality.
- Such traffic can be easily intercepted or monitored by attackers on the same network.

2.5 HTTPS Traffic Analysis (Encrypted Communication)

HTTPS traffic was analyzed to observe how encryption protects data during transmission.

Action Performed

- The website <https://www.google.com> was accessed.
- The Wireshark display filter `tls` was applied.

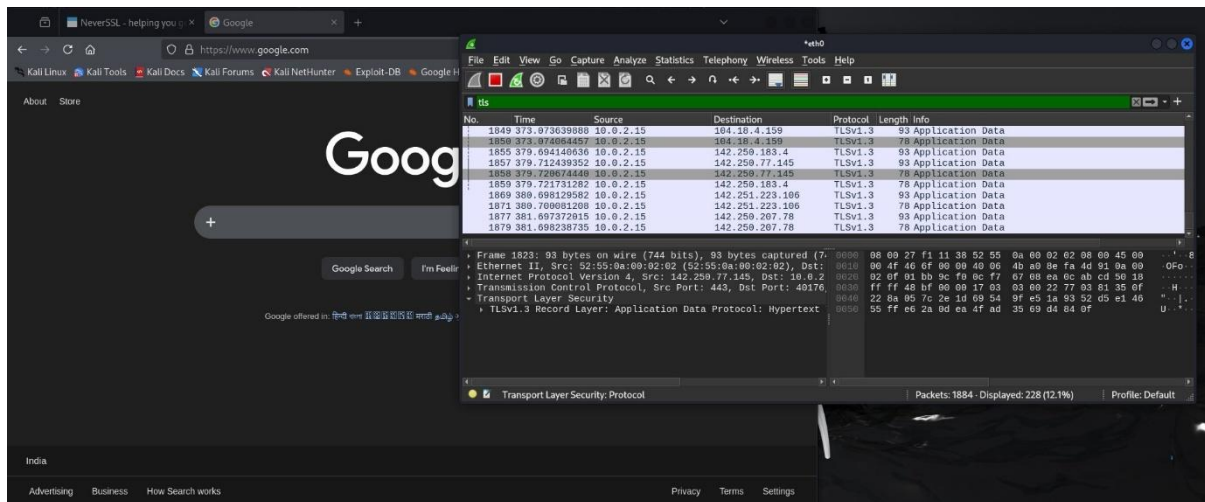


Figure 4: HTTPS Traffic Encrypted Using TLS Protocol

Observation

- The captured packets were identified as TLS traffic.
- Application data appeared encrypted and unreadable.
- Sensitive information such as URLs and headers was not visible.
- This shows how HTTPS protects data from network sniffing attacks.

2.6 TCP Three-Way Handshake Analysis

The TCP connection establishment process was analyzed to understand how reliable communication is initiated.

Action Performed

- Network activity was generated by accessing a website.
- The Wireshark display filter “tcp.flags.syn == 1” was applied.

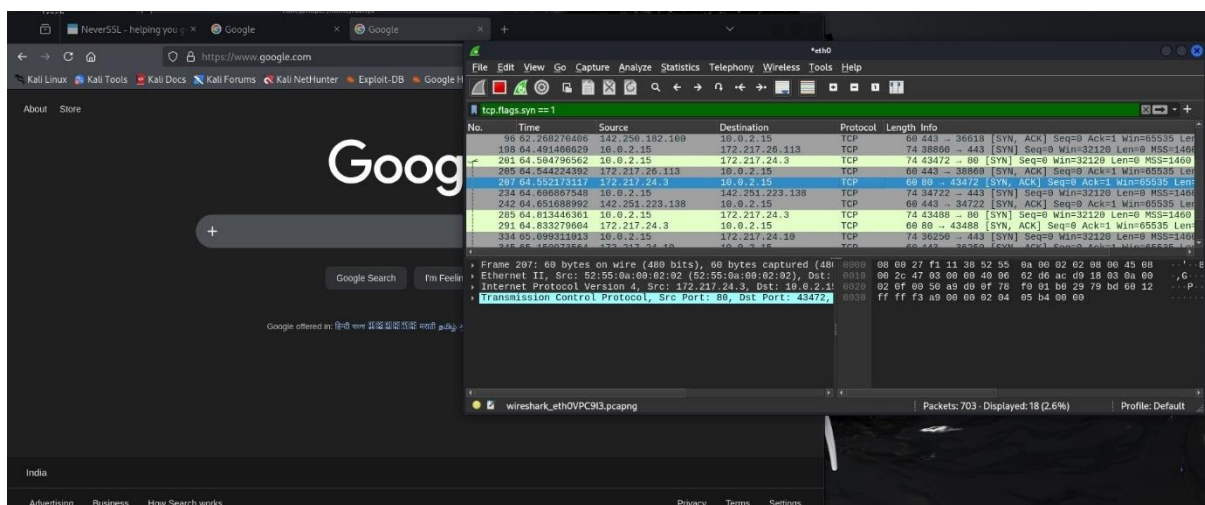


Figure 5: TCP Three-Way Handshake Observed in Wireshark

Observation

- A SYN packet was sent by the client to initiate the connection.
- The server responded with a SYN-ACK packet.
- The client completed the process by sending an ACK packet.
- This three-step process ensures a reliable connection before data transfer begins.

3.1 Overall Observations from Network Traffic

By using Wireshark to capture live network traffic, it became clear that **every action performed on a system creates network activity**. Simple actions such as pinging a server, resolving a domain name, or opening a website all generate packets that travel across the network.

This helped in understanding that:

- Network communication happens continuously in the background
- Different actions use different protocols
- Network traffic can be observed and analyzed if not properly protected

3.2 Security Understanding from ICMP Traffic

From the ICMP analysis:

- Ping requests were sent from the local system
- Ping replies were received from the remote server

From a security point of view:

- ICMP is useful for checking connectivity
- Attackers can also misuse ICMP to scan systems or flood networks
- Monitoring ICMP traffic helps detect unusual behaviour

Even simple protocols like ICMP can become security risks if not controlled.

3.3 Security Understanding from DNS Traffic

DNS analysis showed that:

- Domain names must be converted into IP addresses before communication
- DNS queries and responses are visible on the network

From a cybersecurity perspective:

- DNS traffic can reveal which websites are being accessed
- Attackers can use malicious domains to communicate with infected systems
- Monitoring DNS activity helps detect suspicious or unsafe domains

This highlights the importance of securing and monitoring DNS traffic.

3.4 Security Difference Between HTTP and HTTPS

The comparison between HTTP and HTTPS clearly showed how encryption affects security.

What was observed:

- HTTP traffic displayed readable request information
- HTTPS traffic appeared encrypted and unreadable

Security meaning:

- HTTP does not protect data and is vulnerable to interception
- HTTPS encrypts data using TLS
- Encryption prevents attackers from reading sensitive information

This practical observation made it clear why HTTPS is essential for secure communication.

3.5 Security Insight from TCP Three-Way Handshake

The TCP three-way handshake showed how a reliable connection is established before data transfer.

From a security perspective:

- TCP ensures both systems agree before communication starts
- Abnormal connection patterns can indicate scanning or attack attempts
- Understanding TCP behavior helps identify suspicious connections

This step showed how attackers and defenders both rely on TCP behavior.

3.6 Understanding Network Layers Through Practical Observation

Although this task focused on practical work, the captured traffic helped in understanding how different parts of networking work together:

- ICMP and IP-related traffic handle connectivity
- TCP manages reliable data transfer
- HTTP and HTTPS handle application-level communication
- TLS encrypts application data during transmission

This layered view helps security professionals know where to apply monitoring and protection.

3.7 Importance of Packet Analysis in Cybersecurity

Packet analysis is important in cybersecurity because it helps:

- Identify unencrypted or risky traffic
- Detect suspicious network activity
- Understand how attacks move through networks
- Investigate security incidents

Wireshark proved to be a powerful tool for observing and understanding real network behavior.

3.8 Final Conclusion

This task helped in understanding how basic networking concepts work in real scenarios. By analyzing ICMP, DNS, HTTP, HTTPS, and TCP traffic, it became easier to see how data travels across a network and how security mechanisms protect that data.

Observing real network traffic instead of only reading theory made the learning more practical and meaningful. This task reinforced the importance of network monitoring, encryption, and protocol awareness in cybersecurity.